

Major Project (ITM 701)

Project Proposal

Equilibrium.ai

B. E. Information Technology

By

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**University of Mumbai
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Title of the Project: Equilibrium.ai

Domain:AI-ML

I. Introduction

Cognitive load, the mental effort required to process and respond to information, directly affects focus, accuracy, and well-being in today's fast-paced corporate environments. Employees and managers routinely face information overload, digital interruptions, and intensive multitasking, all of which can reduce productivity and lead to burnout if unmanaged. Equilibrium.ai addresses this challenge by providing real-time, AI-driven monitoring of employee cognitive state through webcam-based micro-expression analysis, speech signals, and task context. The platform proactively intervenes with automated adjustments like silencing unnecessary notifications, optimizing workspace settings, suggesting timely breaks, and dynamically prioritizing tasks while preserving individual privacy. Beyond stress detection, Equilibrium.ai empowers users with context-aware tools: adaptive focus modes to minimize interruptions during deep work, smart break scheduling based on actual mental fatigue, meeting and workload management aligned to cognitive capacity, and workspace automation to support sustained productivity. By integrating these features, Equilibrium.ai enables organizations to boost well-being, prevent burnout, and foster a healthier, more productive work culture through timely, intelligent, and personalized support.

II. Motivation

Employees in today's knowledge-driven workplaces face increasing cognitive overload and work-related stress that quietly harm morale, focus, and long-term health. With faster, more complex digital workflows and remote or hybrid setups, early signs of strain often go unnoticed until productivity or well-being suffers. Traditional self-reporting is biased and rarely provides real-time, actionable insights. This project addresses the urgent need for smarter, real-time tools that automatically detect early overload or disengagement and support employees throughout the day. By leveraging AI, automation, and analytics, we can move beyond periodic check-ins and help organizations foster sustained well-being and performance.

III. Problem Statement

To address the growing challenge of cognitive overload and declining well-being in modern workplaces, this project aims to develop an AI-powered framework that continuously and unobtrusively monitors employees' emotional and cognitive states using computer vision and audio analysis. By detecting real-time indicators of stress, fatigue, and disengagement from facial expressions, voice, and work context, the system provides privacy-preserving, actionable insights offering employees personalized reminders for breaks or focus sessions, while enabling managers and HR to view emotional trends through intuitive dashboards.

IV. Literature Review

Sr. No.	Year & Title	Problem Statement	Methodology	Results and Conclusion	Strengths
[1]	Deep-Learning-Based Stress Recognition with Spatial-Temporal Facial Information (2021)	This paper introduces a specialized public video dataset and a state-of-the-art deep-learning model to overcome the weaknesses of handcrafted features and the data scarcity in stress detection.	The Yonsei Stress Image Database: a large-scale public video dataset labeled with three stress levels—neutral, low, and high. A dual-input network that feeds face video frames through ResNet-18.	This model achieves 66.84% accuracy on the Yonsei Stress Image Database - outperforming all baselines - and proves that combining facial images, landmarks, and spatiotemporal attention effectively detects subtle stress cues.	- Fills dataset gap with a large public database - Combines image and landmark streams for richer features. - Attention modules focus on the most informative data.
[2]	Predicting Depression, Anxiety, and Stress Levels from Videos Using FACS(2019)	Fast, non-intrusive prediction of clinical DASS scores from video via FACS rather than questionnaires.	AU Classification: AAM extracts 72 parameters; multiclass SVMs classify intensities of 31 AUs. Feature Matrix: Normalizes those 31 AU intensities into a per-frame “Facial Matrix.” Prediction: FFNN ingests 30-frame matrices to predict depression, anxiety, and stress together.	Stress 90.2%, Depression 87.2% , Anxiety 77.9%; runs fast enough for real-time use.	- First direct FACS→DASS mapping with a three-stage pipeline. - Excellent accuracy and real-time performance.
[3]	(2021) Stress Recognition Using Facial Landmarks and CNN (AlexNet)	Detect cognitive stress non-invasively from ordinary camera images by leveraging facial landmarks instead of bio-signals or thermal imaging	- Extract 64 facial landmarks and crop the face. - Feed landmarks and face crops into an AlexNet-based CNN. - Classify into three stress levels (none, weak, strong).	Achieved 89% accuracy on emotion classification, enabling reliable inference of stress levels consistent with psychological theory	- Uses standard cameras (non-invasive). - Landmark features capture subtle facial cues. - Ready for real-time implementation.

[4]	(2014) Thermal Spatio-Temporal Data for Stress Recognition	Develop a non-intrusive stress detector using both thermal (TS) and visible (VS) face videos, addressing limits of physiological sensors	- Recorded thermal and visible face videos in ANUStressDB. - Extracted LBP-TOP and HDTP features. - Classified three stress levels with GA-optimized SVM.	Fusion of HDTP+LBP-TOP on TS+VS with GA-SVM achieved 86% accuracy, significantly outperforming single-modality baselines	- Leverages thermal+visible data for richer cues. - HDTP normalizes individual thermal baselines. - GA-SVM selects only the most informative facial regions.
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V. Gap Analysis

Research shows a significant gap between academic stress detection and its real-world use in corporate settings. While studies prove stress can be recognized from facial data using models like HMMs and CNNs, these are done in controlled labs with ideal conditions such as frontal camera angles and short-term stressors. The challenge remains how to monitor stress continuously in dynamic offices with varying lighting, head poses, and real demands. This project addresses that by developing a longitudinal system to track stress trends over time and identify burnout risks. Beyond detection, it integrates data into a management ecosystem with a manager's dashboard, team analytics, and actionable interventions like smart notifications and counseling, bridging the gap between identifying and managing stress at work.

VI. Proposed Solution/Methodology

The project is an AI-powered emotional monitoring framework designed to analyze and support employee well-being in real time within office settings. The system aims to detect emotional cues such as stress, tension, fatigue, and engagement levels during daily work and meetings.

Key Components:

- Facial Emotion Recognition Module
 - Uses computer vision techniques (CNN, LSTM) to analyze facial expressions from webcam feeds.
 - Detects subtle muscle movements indicating stress, frustration, or anxiety during work or meetings using facial landmarks or facial feature extraction.
 - Train model to classify stressful facial cues (SVM or CNN).
- Speech and Audio Stress Analysis Module
 - Captures audio from meetings or work sessions.
 - Analyzes voice features like pitch, tone, speech rate, and pauses using MFCC, RNN, and LSTM.
- Real-Time Notification and Intervention Engine
 - Detects high stress or emotional fatigue in employees.

- Sends personalized reminders for breaks or “focus mode” to reduce distractions and prevent burnout.
- Manager Dashboard and Analytics
 - Provides aggregated emotional trends and stress levels for teams.
 - Displays real-time emotion heatmaps, historical graphs, and alerts for sustained stress.
 - Helps managers make informed decisions on workload and employee support.
 - Monitoring employee productivity based on URL tracking and periodical work screenshot monitoring.

VII. Hardware & Software Requirements

Hardware

- CPU: Intel i5 (10th Gen+) / AMD Ryzen 5
- RAM: 16 GB
- Storage: 500 GB+ NVMe SSD
- Webcam
- Microphone

Software

- Programming Languages: Python, JavaScript, Node.js
- Computer Vision/ Image Processing: OpenCV, MediaPipe
- Machine Learning/ Deep Learning Frameworks: TensorFlow, PyTorch, scikit-learn
- Audio/ Speech Processing: Librosa, SpeechRecognition
- Web Frameworks/ Libraries: Flask/ FastAPI (Backend / API), React.js (Frontend)
- Database/ Storage: MongoDB/PostgreSQL, SQLite

VIII. References

[1] T. Jeon, H. B. Bae, Y. Lee, S. Jang, and S. Lee, "Deep-Learning-Based Stress Recognition with Spatial-Temporal Facial Information," *Sensors*, vol. 21, no. 22, Art. no. 7498, Nov. 2021. doi: [10.3390/s21227498](https://doi.org/10.3390/s21227498).

[2] M. Gavrilescu and N. Vizireanu, "Predicting Depression, Anxiety, and Stress Levels from Videos Using the Facial Action Coding System," *Sensors*, vol. 19, no. 17, Art. no. 3693, Aug. 2019. doi: [10.3390/s19173693](https://doi.org/10.3390/s19173693).

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[4] N. Sharma, A. Dhall, T. Gedeon, et al., "Thermal spatio-temporal data for stress recognition," *J. Image Video Process.*, vol. 2014, no. 28, 2014. doi: [10.1186/1687-5281-2014-28](https://doi.org/10.1186/1687-5281-2014-28).